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## 1. MOTIVATION

- XR tours are increasingly used in museums, education, tourism, and cultural heritage.
- Many existing XR tours still rely on manual 3D authoring or fixed-view 360° media.
- These approaches are often costly to produce or restrict free exploration.
- 3D Gaussian Splatting enables photoreal free-view rendering from captured imagery.
- This work presents a guided XR tour built on captured 3DGS scenes

## 2. CONTRIBUTIONS

- A complete XR touring system for free-view exploring captured real-world environments using 3DGS.
- An in-scene avatar guidance method based on proximity-triggered text-to-speech narration.
- A public walk-up case study with more than 50 participants.

## 3. SYSTEM OVERVIEW

- Six indoor and outdoor locations were captured as tour scenes (~500 images each).
- Each scene was reconstructed using a standard 3DGS pipeline.
- Gaussian primitives were streamed into Unity for VR visualisation.
- An avatar was embedded in each scene to provide audio-based contextual guidance.
- The experience was rendered on Meta Quest 3 via Link.
- An in-headset interface (Figure 3) supported switching between scenes.

## 4. SYSTEM OVERVIEW

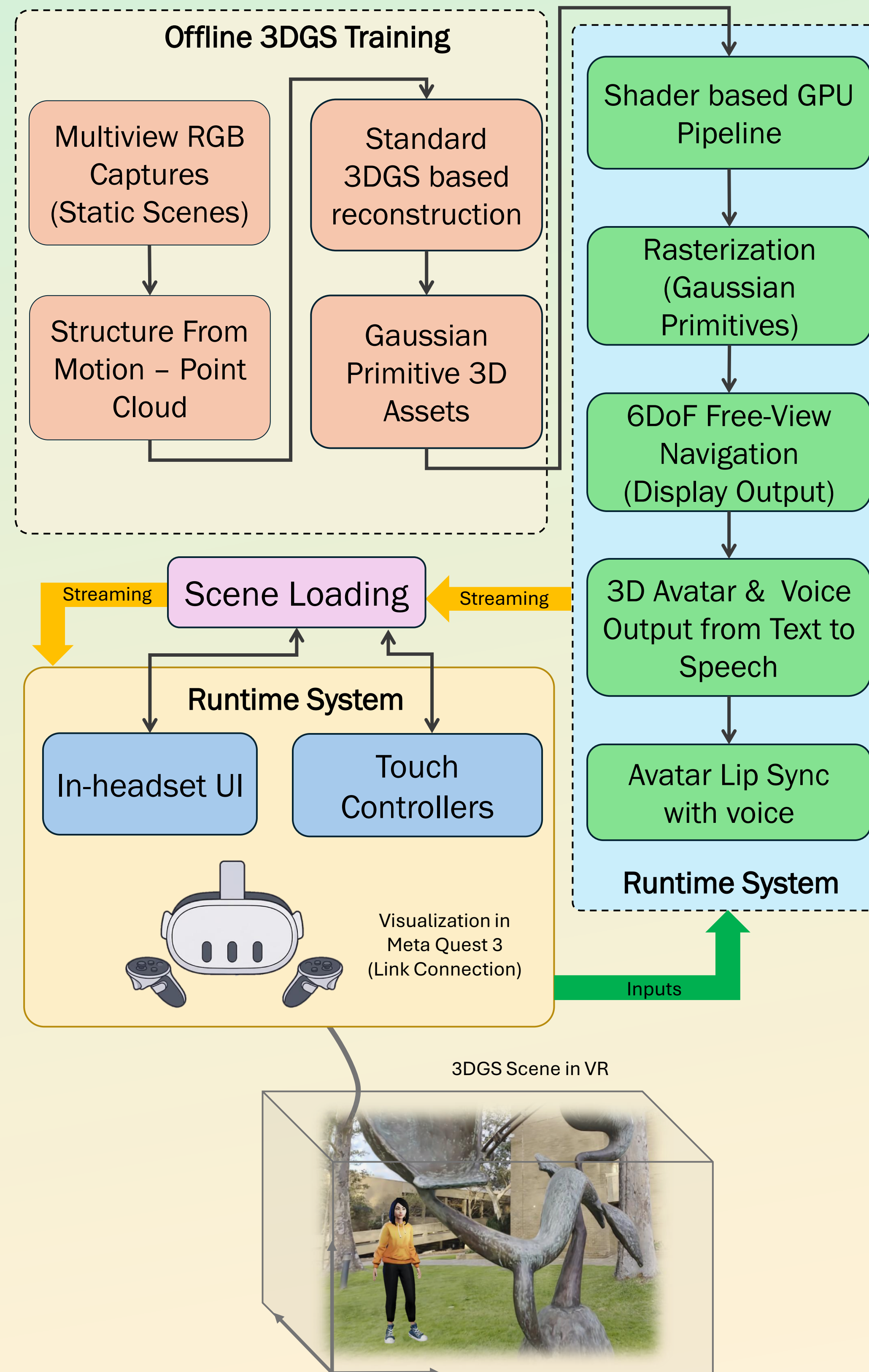


Figure 2 : System Overview

## 5. FREE-VIEW NAVIGATION

- Users could look around and move freely around each captured environment.
- Touch Controllers were used for viewpoint translation and yaw rotation.
- Head pose controlled the viewing direction in VR.
- This design supported self-directed exploration and telepresence.

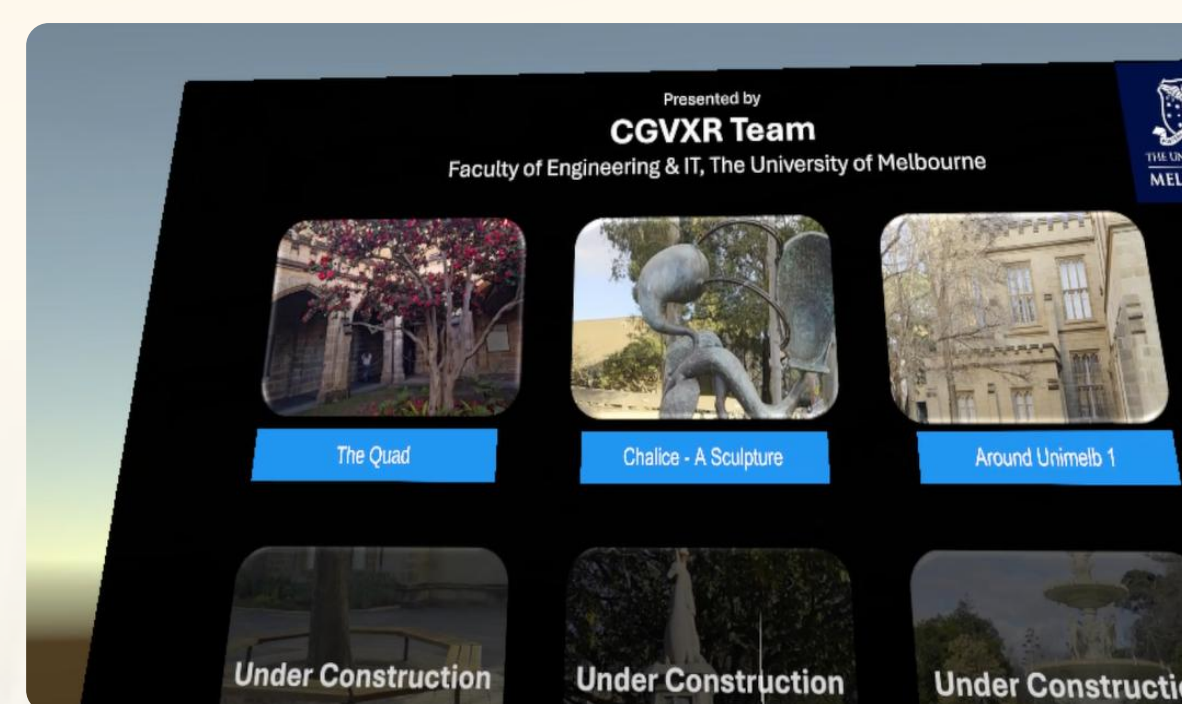


Figure 3 : In headset interface to switch scenes

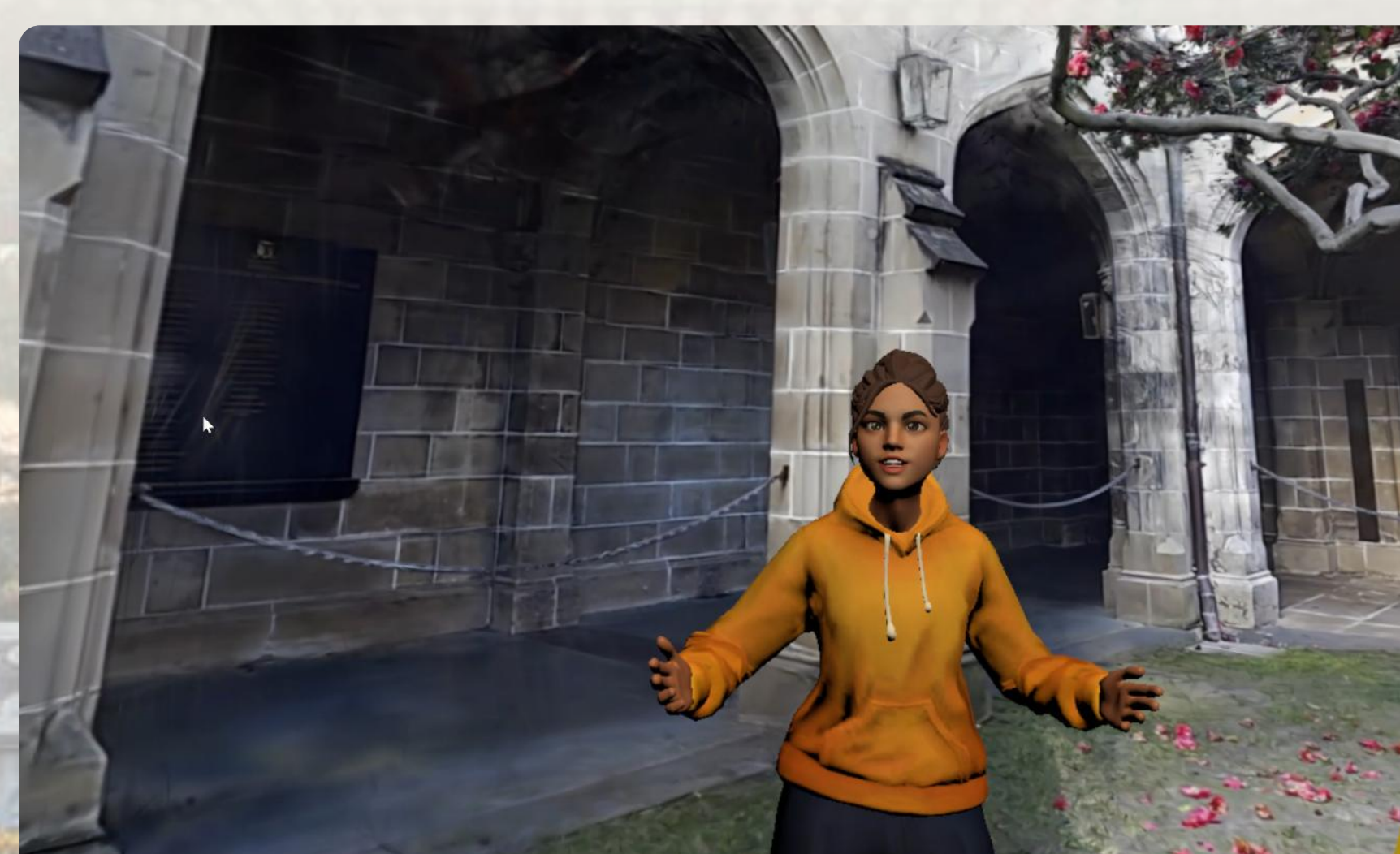


Figure 4 : VR Experience

## 6. AVATAR-GUIDED NARRATION

- A 3D avatar was placed inside each scene as a spatial cue.
- Avatars were designed using *Ready Play Me* [2]
- Narration was triggered when users entered within a proximity radius.
- Text-to-speech audio delivered place-specific information during exploration.
- *Oculus LipSync for Unity* [3] was used for to synchronize mouth and lip movement from narration audio.



Figure 5 : Avatar Embedded in 3DGS Scene

## 7. CASE STUDY

- The system was demonstrated during a public event with over 50 participants.
- Participants commonly reported a strong sense of immersion and telepresence in VR.
- Familiar places felt visually recognizable to previous visitors.
- Free-view navigation gave users agency, while avatar narration added context.



Figure 6 : Case Study presented in a public event

## 8. LIMITATIONS & FUTURE WORK

- 3DGS reconstructions still show visual artefacts.
- Evaluation was limited to public walk-up observations.
- Future work will improve reconstruction quality and scale to larger environments.
- Future work will include controlled user studies.

## REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*,
- [2] Oculus lipsync for unity. <https://developers.meta.com/horizon/documentation/unity/audio-ovrlipsync-unity/>.
- [3] Readyplayer medocumentation. <https://docs.readyplayer.me/>.

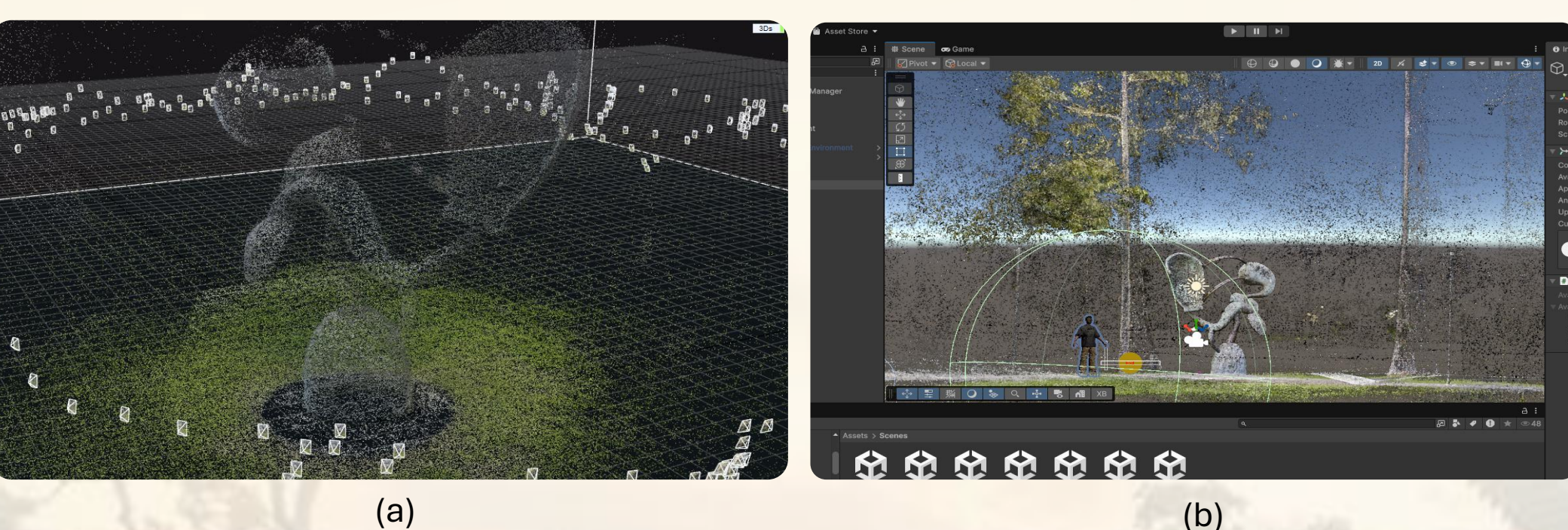


Figure 1 : (a) Sfm Point Clouds; (b) 3DGS Scene in Unity



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